9.6 Making Inferences from Experiments

Name _____ Period ____ Date _____

Algebra II 9.6 · Student sheet

I can…

- find the experimental difference of the means.
- use resampling to test whether a treatment had an effect.
- count the possible resamplings.

Vocabulary · fill in the missing word

Experimental difference — $\bar{x}$(_____) − $\bar{x}$(control), the difference of the sample means

Null hypothesis — the claim that the _____ has no effect

Resampling — shuffling all the data into new groups many times to see what _____ happen by chance

Number of resamplings — ${}_nC_r$ ways to _____ which values go in one group

Part 1 · Shuffle test

Plant growth (cm). Control: 3, 5, 4, 6, 2. With fertilizer: 7, 5, 8, 6, 9.

Observed difference: mean (fertilizer) − mean (control) = _____ − _____ = _____

Shuffle all 10 cards, deal two piles of 5, and find the difference of the means. Repeat 5 times.

Shuffle	1	2	3	4	5
Difference					

How many shuffles (yours and the class's) gave a difference of 3 or more? _____ What does that suggest?

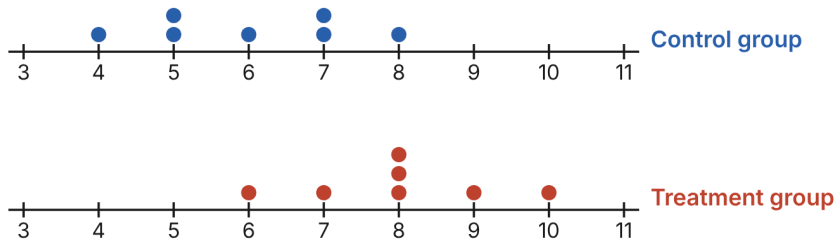
_____

Part 2 · Notes and examples (fill in each note, then work the example)

Experimental difference: Experimental difference = _____ − _____

Example 1

The dot plot shows the results of an experiment. Find the experimental difference of the means. What does it mean?



Resampling: If differences that large are _____ when shuffled, reject the null hypothesis.

Example 2

The experiment in Example 1 was resampled 200 times. Only 6 of the 200 resampled differences were 2 or greater. Should you reject the hypothesis that the treatment has no effect?

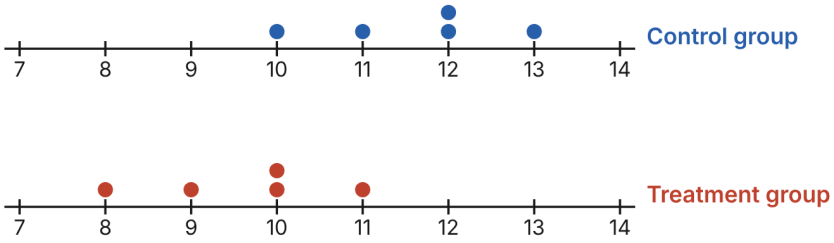
Number of resamplings: Ways to split n values into a group of r: _____

Example 3

How many ways can the 14 values in Example 1 be divided into two groups of 7?

Example 4

Is the experimental difference of the means positive or negative? Explain.



Part 3 · Partner practice

Partner 1

Control: 12, 15, 11, 14. Treatment: 16, 18, 15, 19. Find the experimental difference of the means.

Partner 2

How many ways can 10 values be divided into two groups of 5?